

HeterRAG: Heterogeneous Processing-in-Memory Acceleration for Retrieval-augmented Generation

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Abstract

By integrating external knowledge bases, *Retrieval-augmented Generation* (RAG) enhances natural language generation for knowledge-intensive scenarios and specialized domains, producing content that is both more informative and personalized. RAG systems typically consist of two fundamental stages: retrieval and generation. The retrieval stage experiences low bandwidth utilization due to its random and irregular memory access patterns. Meanwhile, the generation stage is also constrained by memory bandwidth limitations, which arise from involving a significant number of *General Matrix-Vector Multiplications* (GEMV) operations. These two stages collectively lead to memory bottlenecks within RAG systems. Recent efforts leverage HBM-based *Processing-in-Memory* (PIM) to accelerate conventional *Large Language Models* (LLMs). However, the retrieval stage incurs substantial storage overhead due to the need to maintain large-scale knowledge bases, resulting in a capacity bottleneck. Solely relying on HBM-based PIM in RAG is both costly and insufficient to meet the capacity demands. Fortunately, DIMM-based PIM provides a low-cost, high-capacity alternative that complements HBM. In this work, we propose HeterRAG, a novel heterogeneous PIM acceleration system for RAG. It combines

HBM-based PIM and DIMM-based PIM to achieve high performance, energy efficiency, and low hardware cost. HeterRAG uses HBM-based PIM for the generation stage to meet bandwidth needs and DIMM-based PIM for the retrieval stage to satisfy memory capacity requirements. To further improve performance, HeterRAG incorporates three software-hardware co-optimization techniques: locality-aware retrieval, locality-aware generation, and fine-grained parallel pipelining. Experimental results demonstrate that, compared to RAG systems deployed on Intel Xeon CPUs and NVIDIA GPUs, HeterRAG achieves up to 26.5× higher throughput, up to 27.6× lower latency, and up to 2.8× greater energy efficiency.

CCS Concepts

• **Hardware** → **Emerging technologies**; • **Computer systems organization** → **Parallel architectures**.

Keywords

Retrieval-augmented generation, Large language models, Approximate nearest neighbor search, Processing-in-memory, DIMM, HBM

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1 Introduction

In recent years, *Large Language Models* (LLMs) have made significant advancements in the field of natural language processing [27, 79, 90]. With their powerful generative capabilities, LLMs can produce high-quality text across various tasks. However, LLMs

rely on pretraining data and cannot provide accurate real-time responses [28]. To address this limitation, the *Retrieval-augmented Generation* (RAG) method has emerged [7, 59, 68, 74]. RAG combines a knowledge base and retrieves relevant documents to support generation in real-time, enhancing the performance of LLMs in complex information-demanding contexts [40, 72]. Beyond integrating general knowledge bases, RAG can also construct domain-specific knowledge bases. For example, deploying private RAG systems within enterprises has become a trend [2, 20, 55, 80]. By integrating large models with private knowledge bases, enterprises can fully leverage their internal data while ensuring data privacy, thereby improving decision support and information generation efficiency.

RAG systems consist of two main stages: retrieval and generation. In the retrieval stage, the system efficiently retrieves relevant documents through *Approximate Nearest Neighbor Search* (ANNS) [48, 52, 84], providing accurate contextual support for generation. A commonly adopted approach is the graph-based ANNS [19, 58, 82], which utilizes a pre-built graph index to match query vectors with document vectors. Graph-based ANNS can quickly identify the most similar document set, thereby significantly improving retrieval efficiency. Then, in the generation stage, the system augments the LLM's input with retrieved documents to generate more accurate and relevant responses.

Although RAG offers significant advantages over traditional LLMs, it also faces two major system-level challenges. First, RAG systems require high memory bandwidth. In the retrieval stage, the system searches for relevant documents through a graph index. This involves random and irregular memory access due to the sparsity of the graph and the randomness of search paths, resulting in low bandwidth utilization. In the generation stage, the transformer's autoregressive decoding and attention mechanism [79] requires many *General Matrix-Vector Multiplications* (GEMV) operations. Limited reuse of weights and *Key-Value* (KV) cache causes frequent data loading from off-chip memory, making performance highly dependent on memory bandwidth. Second, RAG systems demand a large memory capacity. In addition to the memory required for the generation stage, the retrieval stage relies on large knowledge bases containing massive documents, vectors, and indexes, which demands significant additional memory capacity. With the continuous growth of real-world knowledge, the memory requirements of RAG systems are expected to keep increasing.

Memory-bound applications often use *Processing-in-Memory* (PIM) technology [44, 61] to improve performance. Currently, most approaches rely on HBM-based PIM to accelerate large model inference because of HBM's high bandwidth and low power consumption [26, 64, 93]. However, in RAG systems, using only HBM-based PIM leads to high hardware costs. For example, A single HBM stack typically provides up to 24GB of capacity and can cost approximately five times more per gigabyte than DDR5 DRAM [83]. Moreover, HBM prices are expected to rise due to supply constraints and increasing demand from artificial intelligence and data center applications [77]. Meeting RAG's memory capacity needs would require many HBM stacks, significantly increasing system cost. On the other hand, DDR-based DIMMs can reach a capacity of 64GB [32], making them much cheaper and more spacious than HBM. A naïve solution is to combine DIMM with HBM-based PIM to store the large knowledge base. However, both retrieval and

generation stages still need to transfer data to the HBM-based PIM for processing, causing heavy data transfer overhead. RAG systems become less efficient because the bandwidth of interconnects like PCIe is significantly lower than the internal memory bandwidth.

In this paper, we introduce HeterRAG, a heterogeneous PIM system that accelerates RAG by combining the strengths of both HBM-based PIM and DIMM-based PIM. For the generation, we follow the current HBM-based PIM acceleration design for LLM inference to leverage HBM's high bandwidth and low power consumption. For the retrieval, which has higher memory demands, we use DIMM-based PIM to benefit from DIMM's large capacity and low cost. The HBM-based PIM and DIMM-based PIM are designed as independent devices that can be scaled and optimized separately to better accommodate diverse RAG system workloads.

HeterRAG also adopts three software-hardware co-optimization techniques: locality-aware retrieval, locality-aware generation, and fine-grained parallel pipeline. In the retrieval stage, we cache frequently accessed vertex vectors to reduce DRAM access and reuse results from previous iterations to accelerate the retrieval process. In the generation stage, we cache the frequently accessed KV tensors associated with documents to avoid redundant computations. Additionally, as soon as part of the retrieval stage is complete, we start the generation stage to improve overall hardware utilization.

In summary, we make the following contributions:

- From a computer systems architecture perspective, we examine the emerging RAG domain, identifying that RAG systems face more severe challenges in memory bandwidth and capacity compared to traditional LLMs.
- We propose a heterogeneous PIM system, HeterRAG, which combines the advantages of HBM and DIMM to achieve high performance, energy efficiency, and low cost. HeterRAG primarily includes two heterogeneous PIM devices, AccelDIMM and AccelHBM, which efficiently handle the retrieval and generation parts of RAG systems, respectively.
- Moreover, HeterRAG provides significant benefits through its software-hardware co-optimizations, including locality-aware retrieval, locality-aware generation, and fine-grained parallel pipeline.
- We evaluate HeterRAG on various configurations and demonstrate that it significantly outperforms the CPU-GPU, the naïve HBM-only PIM, and the DIMM-only PIM baselines, achieving a throughput improvement of 6.6× to 26.5× (13.5× on average), a latency reduction of 8.7× to 27.6× (13.7× on average), and significant energy efficiency gains.

2 Background

2.1 Retrieval-augmented Generation

Retrieval-augmented Generation (RAG) is a method that combines information retrieval with natural language generation [40, 59, 72]. By integrating specialized and up-to-date knowledge bases, RAG enables *Large Language Models* (LLMs) to generate responses that are more timely, accurate, and informative.

As shown in Figure 1(a), RAG uses an embedding model during the offline preprocess to transform knowledge (e.g., document chunks) into high-dimensional vectors and construct a specific

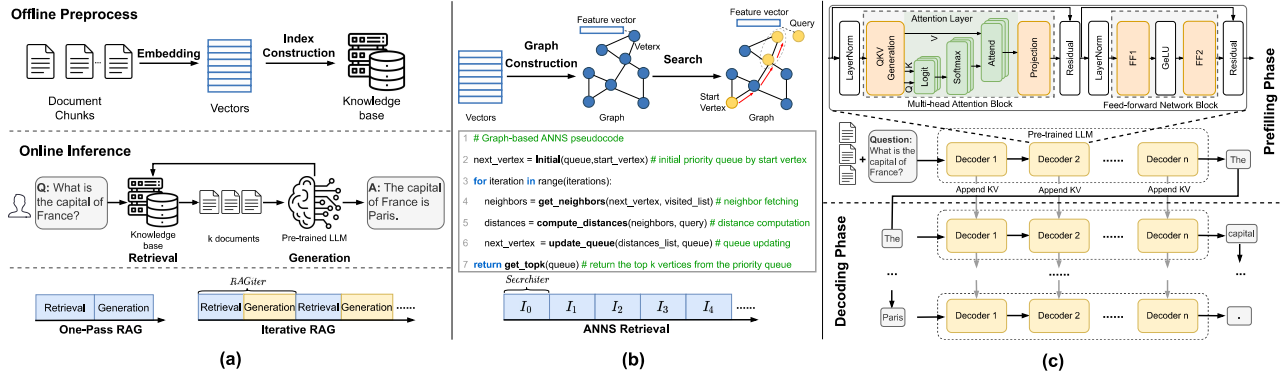


Figure 1: (a) RAG process overview, (b) Graph-based ANNS, and (c) Overview of GPT's architecture and its autoregressive inference process

index, forming a complete knowledge base with the original knowledge. During online inference, the RAG system operates in two stages: retrieval and generation. It first retrieves relevant information from the knowledge base according to the user request, then uses a pre-trained large model to generate a response.

Currently, RAG operates in two main modes: one-pass RAG and iterative RAG. One-pass RAG [66, 81] performs a one-time retrieval and generation, where the system retrieves relevant documents from the knowledge base based on the user request and directly generates a response by inputting these documents along with the request into the model. Iterative RAG [7, 48, 68, 78], in contrast, involves multiple rounds of retrieval and generation, where the system may refine the query at each iteration, progressively retrieving more relevant information to produce more accurate and contextually appropriate answers.

Retrieval Stage: RAG converts user requests or the latest context into high-dimensional vectors and uses them as queries to search the pre-constructed knowledge base [54], retrieving semantically relevant documents. To accelerate large-scale vector retrieval, *Approximate Nearest Neighbor Search* (ANNS) [52] techniques have been widely adopted, with graph-based ANNS [19, 58] emerging as a leading approach due to its high speed and accuracy.

Graph-based ANNS represents vectors as vertices and organizes them into a graph structure [53]. During retrieval, it employs a best-first search strategy [82], iterating multiple times to identify the most relevant results as illustrated in Figure 1(b). The process begins with initializing a priority queue with the starting vertices. In each iteration, graph-based ANNS selects the nearest unchecked vertex from the queue. It then fetches its neighbors, filters out previously visited ones to avoid redundancy (*neighbor fetching*), computes the distances between the query vector and the filtered neighbors (*distance computation*), and updates the queue with the new distances (*queue updating*).

Generation Stage: RAG's generation is typically driven by LLMs built on decoder-only transformer architectures [90], such as GPT [8] and LLaMa [75]. As shown in Figure 1(c), GPT is composed of a stack of decoder blocks. Each block contains a *Multi-head Attention* (MHA) module and a *Feedforward Network* (FFN) module, both preceded by *Layer Normalization* (LayerNorm) and followed by residual connections. The MHA module includes a QKV generation layer, an attention layer, and an output projection layer,

while the FFN module comprises two fully connected layers (FF1 and FF2) and a *Gaussian Error Linear Unit* (GELU) activation. In contrast, LLaMa replaces the GELU activation with *SwiGLU* in the FFN module and employs *Rotary Positional Embeddings* (RoPE) for position encoding.

The inference process of LLMs typically consists of two sub-phases: the *prefilling* phase and the *decoding* phase. In the *prefilling* phase, the entire token sequence is processed in parallel at the token level, with the dominant computation being *General Matrix-Matrix Multiplication* (GEMM). In contrast, the *decoding* phase proceeds autoregressively, generating one token at a time based on previously generated tokens, where the computation is typically dominated by *General Matrix-Vector Multiplication* (GEMV) due to the single-token input.

2.2 Processing-in-Memory

Traditional von Neumann architectures separate computation and storage, requiring frequent data transfers between memory and processors, making the system performance limited by latency and bandwidth. *Processing-in-Memory* (PIM) [13, 44, 46, 56, 70] technology addresses this by integrating computation within or near memory, reducing data movement and energy consumption. PIM can be broadly categorized into analog and digital approaches. Analog PIM, such as ReRAM-based in-memory computing [29, 30], performs operations using physical memory properties. Digital PIM employs embedded processing units to execute computations closer to memory [3, 25, 39, 43, 45, 93]. This work focuses on DIMM-based and HBM-based digital PIM, chosen for their popularity, practicality, and reliability.

DIMM-based PIM: DIMMs use a 2D layout of DDR chips, which keeps manufacturing costs low, allowing users to get high-capacity memory at an affordable price. DIMM-based PIM places processing units at different memory levels, such as rank, bank group, and bank levels, to enable efficient data processing. It is widely adopted in applications such as recommendation systems [3, 39, 56, 65] and graph processing [12], where large-scale data is typically involved.

HBM-based PIM: HBM uses a 3D-stacked structure with memory dies vertically connected through *Through Silicon Vias* (TSVs), providing high bandwidth, low latency, and low power consumption. However, its complex manufacturing process leads to high cost

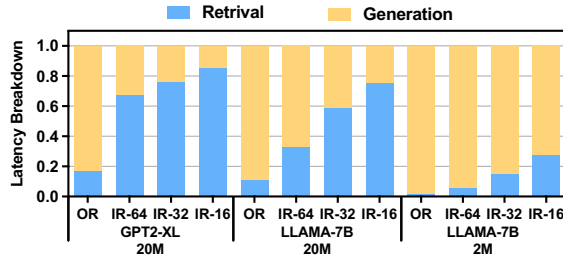


Figure 2: Execution time breakdown of a RAG system using GPT2-XL and LLaMA2-7B models for generation and HNSW for retrieval

and limited per-stack capacity. HBM-based PIM enhances effective bandwidth utilization by integrating processing units near or within memory banks. This makes it ideal for bandwidth-heavy tasks like image processing [21] and machine learning [25, 43, 45, 93].

3 Motivation

In this section, we provide an in-depth analysis of the characterization for both stages of the RAG and motivate why a heterogeneous PIM-accelerated system is needed.

3.1 Characterization of RAG

We analyze the characteristics of each stage on a CPU-GPU-based platform (like the existing RAG systems [34, 35]), where the retrieval stage is performed on the CPU side using the graph-based ANNS algorithm HNSW [58] and the generation stage on the GPU side using GPT2 and LLaMA2. Details about the hardware configuration, knowledge base, and datasets are provided in §5.1.

Execution Time Breakdown: Figure 2 shows the execution time distribution of the RAG system across different knowledge base sizes and models. When the knowledge base is relatively small, the generation stage dominates the execution time. As the size of the knowledge base increases, the retrieval stage needs a significantly higher process time. Additionally, the execution time for both stages varies with the scale of the models and the number of RAG iterations. Therefore, both the retrieval and generation stages can take up considerable execution time in the RAG workflow, indicating that performance optimization should focus on both stages.

Arithmetic Intensity Analysis: To better understand the RAG system, we use the roofline model to analyze its computational and memory access behavior. Figures 3 and 4 show the roofline data points for the retrieval and generation stages, respectively.

Due to its extensive and irregular off-chip memory access, the memory bandwidth becomes the main bottleneck of the retrieval stage. As mentioned earlier, the retrieval stage comprises three operations. Among them, *distance computation* accounts for over 80% of the whole retrieval execution time. The reason is that calculating distances requires accessing sparse graph structures and scattered vector data, making it difficult to predict memory access addresses to prefetch. Moreover, the large size of graph data makes the caching technique less effective.

Similarly, the generation stage is primarily limited by the memory bandwidth. Yet, the two stages' memory access behaviors are quite different. Unlike random, irregular memory access patterns

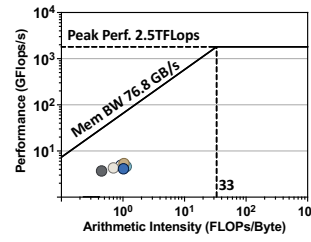


Figure 3: Roofline model analysis of the HNSW algorithm on an Intel Xeon Gold 5117 CPU

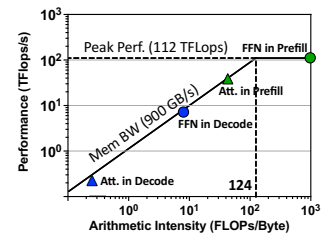


Figure 4: Roofline model analysis of the LLaMA2-7B model layers on an NVIDIA V100 GPU

in the retrieval stage, the weight data from the generation stage is rather regular and sequential. However, the limited on-chip cache capacity cannot accommodate the large weight data together, causing frequent off-chip memory access and thus making the generation stage memory-bound. While batch processing can partially mitigate this limitation, the unique KV cache required by each request still makes the on-chip cache requirement considerable and does not change the memory-bound characteristic. In addition, RAG applications often handle personalized and sensitive user data, and typically cater to a limited number of users, resulting in small batch sizes.

Memory Footprint Analysis: RAG systems rely on knowledge bases for retrieval rather than encoding knowledge in model parameters. This allows smaller models to achieve comparable or even better performance compared to the large models in various knowledge-intensive tasks [23, 31, 48]. The popular models, such as GPT2-XL and LLaMA2-7B, require approximately 3GB and 14GB of memory, respectively, with FP16 precision. For GPT2-XL and LLaMA2-7B, a batch size of 32 with token lengths of 1024 for both the input and output, KV caching requires an additional 18GB and 32GB of memory, respectively. However, RAG systems' memory demands are considerably higher, as their knowledge bases typically store large collections of documents, vectors, and retrieval indices. A knowledge base with tens to hundreds of millions of documents, where each document takes tens to hundreds of bytes, requires a total of hundreds of gigabytes to terabytes of memory. Moreover, as real-world applications continuously incorporate new data, the memory demands of RAG systems steadily increase.

Summary: Random memory access in retrieval and limited data reuse during generation make RAG systems highly dependent on memory bandwidth. Furthermore, the large knowledge base demands high memory capacity.

3.2 Heterogeneous PIM Acceleration for RAG

Why Choose PIM and How to Design It? Analysis of arithmetic intensity shows that increasing memory bandwidth is crucial to improving the performance of RAG systems. PIM technology, which offers higher bandwidth and lower memory access latency, emerges as a promising solution to alleviate memory bottlenecks.

HBM-based PIM is the preferred choice for accelerating large model inference due to its high bandwidth and low per-bit energy consumption. For example, an 8-Hi HBM2 stack (8 dies) provides up to 256GB/s of bandwidth, about 13× higher than the 19.2GB/s

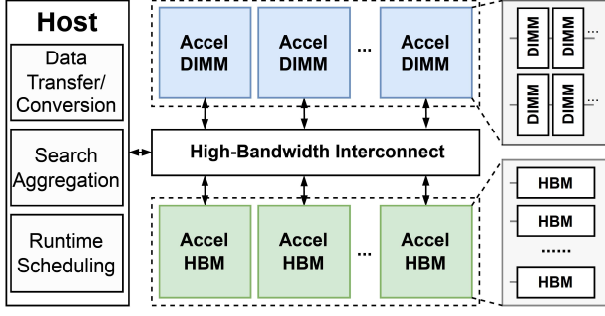


Figure 5: HeterRAG architecture overview

of a DDR4-2400 DIMM, while consuming roughly half the energy per bit transfer [45]. However, for memory-intensive RAG systems, HBM-based PIM is impractical due to its limited capacity and high cost. For instance, an 8-Hi HBM2E stack used in the A100 GPU offers 16GB of capacity at an estimated cost of approximately \$110/GB [62]. Storing even a moderately sized 100GB knowledge base would cost \$11,000, making HBM-based PIM economically impractical for RAG applications.

Compared to HBM, DIMMs offers significant advantages in terms of capacity and cost. For instance, a standard DDR4-based DIMM module can provide up to 64GB of capacity [32], which is 4× that of a single HBM2E stack. Moreover, the simplified manufacturing process of DDR leads to significantly lower costs compared to HBM, with the unit price of even DDR5-based DIMMs being approximately one-fifth that of HBM [76, 83]. Therefore, DIMMs can serve as a simple and cost-effective auxiliary storage in RAG systems. However, if DIMMs are simply combined with HBM-based PIM, all data needs to be transferred from DIMMs to HBM for processing. This transfer is constrained by the limited bandwidth of the interconnect (e.g., PCIe), which is significantly lower than the internal memory bandwidth. As a result, this data movement bottleneck can diminish the performance benefits of PIM, which we will further illustrate in §5.2.

Heterogeneous PIM System: We are therefore motivated to propose a RAG system that ensures efficiency, cost-effectiveness, and scalability by combining the strengths of HBM-based PIM and DIMM-based PIM. HBM-based PIM provides sufficient bandwidth for efficient processing, and DIMM-based PIM offers large storage capacity at low cost, addressing interconnect bottlenecks. This heterogeneous architecture enables independent scaling of DIMM-based and HBM-based PIM, enhancing the system’s flexibility and scalability. However, achieving this goal remains challenging. Firstly, designing distinct PIM architectures to support retrieval and generation tasks introduces complexity. Secondly, RAG systems may suffer from inefficiencies due to the high locality of documents and the interdependence between these two stages. Lastly, managing and unifying heterogeneous PIM devices for user-friendly operation presents an additional challenge. Therefore, effective solutions for heterogeneous PIM-based RAG systems remain lacking.

4 HeterRAG Architecture

In this section, we first present the HeterRAG architecture overview. We then describe the design and functionality of the AccelDIMM

and AccelHBM devices within HeterRAG. Finally, we discuss the software-hardware co-optimizations and the software stack.

4.1 Architecture Overview

Figure 5 shows the proposed HeterRAG architecture, which consists of a host, a high-bandwidth interconnect, and multiple AccelDIMM and AccelHBM devices.

Host: The host primarily manages lightweight tasks during runtime, including data transmission/conversion, scheduling heterogeneous PIM devices, and aggregating retrieval results from AccelDIMM devices. Transferring retrieval results from multiple AccelDIMMs imposes a negligible strain on interconnect bandwidth and host computation. For example, if 10 AccelDIMMs return 100 results each, the total transfer is only 8 KB, and sorting 1000 entries is well within the host CPU’s capability.

Interconnect: In HeterRAG, devices are interconnected through high-bandwidth links, such as *Compute Express Link* (CXL) [16]. HeterRAG allows the number of AccelDIMM and AccelHBM devices to be independently scaled to meet the specific requirements of the RAG system, ensuring scalability. For instance, AccelDIMM and AccelHBM can be implemented as CXL Type 2 devices (similar to GPUs/FPGAs) and connected to a CXL switch via PCIe slots, enabling efficient interconnection. Three CXL sub-protocols are used: *cxl.io*, *cxl.cache*, and *cxl.mem*. The transfer bandwidth between the host and the devices, based on PCIe 5.0 ×8, is approximately 32GB/s. The CXL switch network can support up to 4K nodes [16], offering the potential to meet growing data demands.

AccelDIMM: AccelDIMM is a DIMM-based PIM device specifically designed for the retrieval stage in RAG systems. Multiple AccelDIMM devices operate in parallel to efficiently process large-scale vector data. For offline preprocessing, each AccelDIMM stores a subset of the data and independently constructs graph indices. During online retrieval, query requests are broadcast to all AccelDIMM devices, which independently execute the retrieval tasks. The host then aggregates the results from all devices.

AccelHBM: AccelHBM is an HBM-based PIM device specifically optimized for the generation stage in RAG systems. Scaling the number of AccelHBM devices enables support for larger models. HeterRAG supports both pipeline and tensor parallelism to improve generation efficiency. By default, HeterRAG uses tensor parallelism, where each AccelHBM computes partial results and aggregates them. When the model size is too large, pipeline parallelism is introduced, with each AccelHBM handling a different model stage in a pipelined execution flow. Additionally, users can configure the parallelization strategy based on workload characteristics and system constraints.

Execution Flow: The host receives user queries and converts them into vectors, which are broadcast to all AccelDIMM devices via the interconnect network. The retrieval operations are offloaded to the AccelDIMM devices, where each performs ANNS independently to find the top-*k* nearest vectors to the query. The host then aggregates results from all devices, sorts them, and identifies the final top-*k* nearest vectors. It maps the vectors’ IDs to their corresponding document text, combines it with the user query, and encodes the data into tensors. These tensors are sent to the AccelHBM devices for generation. The AccelHBM devices process the

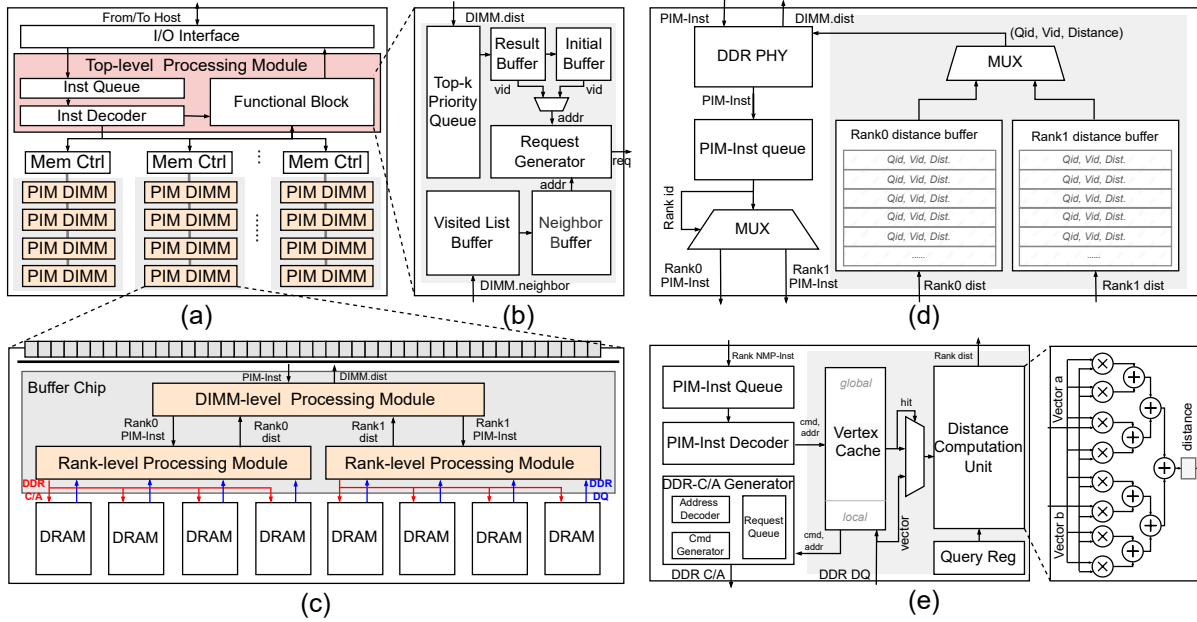


Figure 6: (a) AccelDIMM architecture, (b) Functional block, (c) PIM-enabled DIMM, (d) DIMM-level processing module, and (e) Rank-level processing module

tensors to generate token sequences, which are returned to the host. In iterative RAG, the host derives the next query and repeats the retrieval-generation process until a termination condition is met, such as reaching the maximum iteration count or achieving sufficient confidence. Finally, the host decodes the token sequence into natural language text and returns it to the user as the answer.

4.2 AccelDIMM Architecture

AccelDIMM devices are designed to support completing ANNS operations. *Neighbor fetching* primarily involves memory access and result filtering. However, maintaining a vertex list of access records in memory introduces significant area overhead, making it unsuitable for memory offloading. Similarly, *queue updating*, which requires centralized aggregation of computation results, is also unsuitable for offloading. As a result, memory offloading is restricted to *distance computation* operation.

Figure 6(a) illustrates the architecture design of AccelDIMM. To facilitate interaction with the host, a device interface is implemented to receive host instructions. The top-level processing module is responsible for parsing host instructions, managing device memory, and performing tasks such as *neighbor fetching* and *queue updating*. AccelDIMM features multiple memory channels, with each channel interfacing with PIM-enabled DIMMs through dedicated memory controllers. These PIM-enabled DIMMs are the core components of the system. They adopt a unified memory organization [70] to store vertex neighbors and vector data, and support in-memory distance computation. The system’s capacity can be scaled by adding more DIMMs. The memory controller supports both standard DDR requests and PIM-specific requests. Each AccelDIMM partitions graph data at the vertex level and evenly distributes it across ranks to ensure balanced workload distribution. The following presents the detailed design of each component.

Top-level Processing Module (TPM): The TPM consists of an instruction queue, an instruction decoder, and a functional block. Instructions received from the I/O interface are stored in the instruction queue and decoded by the instruction decoder to initiate ANNS operations on the AccelDIMM. The functional block manages *neighbor fetching* and *queue updating* while generating PIM requests for *distance computation*.

Functional Block (FB): The design of the FB is shown in Figure 6(b). For *neighbor fetching* tasks, the request generator uses vertex IDs from the initial buffer or result buffer as addresses to generate standard DDR read requests. The neighbor data read from memory is filtered through the visited list and stored in the neighbor buffer. The initial buffer contains fixed starting vertices for the search. For *queue updating*, the top-k priority queue receives computation results from the DIMMs, sorts them, and saves the nearest neighbors in the result buffer. For PIM request generation, the request generator uses vertex IDs from the neighbor buffer to generate PIM requests.

Memory Controller: To support PIM requests, we integrate a PIM extension into the memory controller to handle tasks such as address mapping and instruction generation. To address the C/A bandwidth limitations in DIMMs, we employ an instruction compression technique similar to RecNMP [39], embedding all DDR commands for a single PIM request into a single PIM instruction. Additionally, we incorporate the two-phase instruction transmission technique proposed by TRiM [65] to minimize instruction waiting times for the in-memory processing module.

PIM-enabled DIMM: Due to the irregular and sparse memory access patterns of ANNS, deploying *distance computation* at the bank level in DIMMs leads to idle computational resources, higher area overhead, and limited performance gains. To address this, we deploy *distance computation* at the rank level. As illustrated in

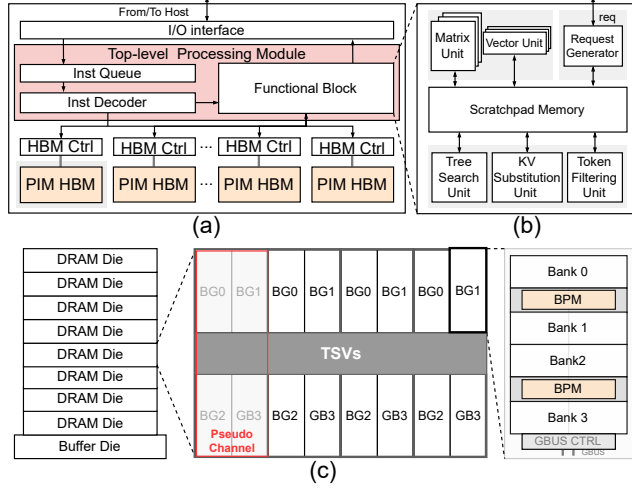


Figure 7: (a) AccelHBM architecture, (b) Functional block, and (c) PIM-enabled HBM

Figure 6(c), two types of processing modules are integrated into the buffer chip: DIMM-level and Rank-level processing modules. Each rank is equipped with a rank-level processing module, enabling multiple ranks to process PIM requests in parallel. For standard DDR commands, the DIMM bypasses these memory processing modules and directly conducts commands such as read and write.

DIMM-level Processing Module (DPM): Figure 6(d) illustrates the design of the DPM. This module receives PIM instructions from the DIMM interface and dispatches them to the corresponding rank-level processing modules based on the Rank ID specified in the instructions. Each rank is allocated a dedicated buffer in the DPM to store computation results from the rank-level modules temporarily. These results are then sent back to higher-level components of AccelDIMM via the standard DIMM interface in a round-robin manner for further processing.

Rank-level Processing Module (RPM): Figure 6(e) illustrates the architecture of the RPM, which performs two key functions: PIM instruction decoding and vector distance computation. This module receives PIM instructions from the DPM and decodes them. The DRAM addr field in the PIM instruction serves as a tag to check the vector cache. In the case of a cache miss, the DRAM addr field and DDR command information fields in the PIM instruction are fed into the DDR-C/A generator to generate standard DDR ACT/RD/PRE commands, which retrieve the corresponding vector data from the DRAM. Vector data, whether from cache or DRAM, is then sent to the distance computation unit to compute its distance to the query vector. The distance computation unit is currently designed to support inner product distance, which is the most commonly used metric in RAG. With minor modifications, it can also be extended to support other distance metrics, such as L2 distance.

4.3 AccelHBM Architecture

We explore accelerating LLMs using PIM at the operation level. LLM inference workloads mainly involve three types of operations: GEMM, GEMV, and other operations such as element-wise computations and normalization (e.g., LayerNorm, RMSNorm, and Softmax).

GEMV, with its low arithmetic intensity, is ideal for in-memory processing and is fully offloaded to memory, while GEMM and other operations are handled outside the memory. Within each AccelHBM, we adopt the same mapping scheme as AttAcc [64] for organizing the KV matrices and weight matrices, ensuring parallelism and minimizing data movement. Based on this design, the architecture of AccelHBM is similar to that of AccelDIMM. As shown in Figure 7(a), its core components include a top-level processing module and PIM-enabled HBM. The following focuses on the key differences between the two architectures.

Top-level Processing Module (TPM): In the TPM, the decoder decodes instructions and triggers LLM generation operations. The *Functional Block* (FB), shown in Figure 7(b), handles all tasks not offloaded to memory. Within the FB, a matrix unit built with a systolic array is dedicated to GEMM computations, while a vector unit composed of multiple *Very Long Instruction Word* (VLIW) processors [17] supports all other operations. A scratchpad memory module provides data support for all computation units. The request generator issues PIM requests to trigger in-memory operations or conventional HBM requests for data read and write. PIM requests can be broadcast to all HBM controllers to enable parallel execution across memory channels. Additionally, the TPM incorporates a tree search unit, a KV substitution unit, and a token filtering unit to support the proposed optimization strategies (detailed in §4.4).

PIM-enabled HBM: The architecture of PIM-enabled HBM is shown in Figure 7(c). It stores weights, KV cache, and other data while supporting in-memory GEMV computations through a unified memory organization. To maximize computational efficiency, we draw inspiration from state-of-the-art GEMV-optimized PIM accelerators, such as Newton [25], and deploy processing modules at the bank level within HBM to enable parallel execution across banks. Each *Bank-level Processing Module* (BPM) includes two computation units for vector inner product operations (similar to the distance computation units in AccelDIMM), with input data sourced from the bank-level row buffer and the channel-level global buffer.

4.4 Software-Hardware Co-optimizations

RAG systems also suffer from several key system-level inefficiencies that are difficult to address through hardware-only solutions.

Firstly, document usage in RAG systems exhibits high locality. On the one hand, documents referenced by different user queries follow a power-law distribution, where most requests concentrate on a small subset of popular documents (e.g., 60% requests are directed toward just the top 3% of documents [35]). On the other hand, in iterative RAG scenarios, user queries often reuse the same documents across multiple iterations. However, current systems do not fully exploit this locality, resulting in redundant data transfers and computations.

Secondly, the retrieval and generation stages in RAG systems are strictly interdependent: the generation stage cannot begin until retrieval is complete. Since these stages are processed by different types of heterogeneous PIM devices (e.g., AccelDIMM and AccelHBM), this sequential execution pattern often leaves one type of device idle, resulting in poor hardware utilization.

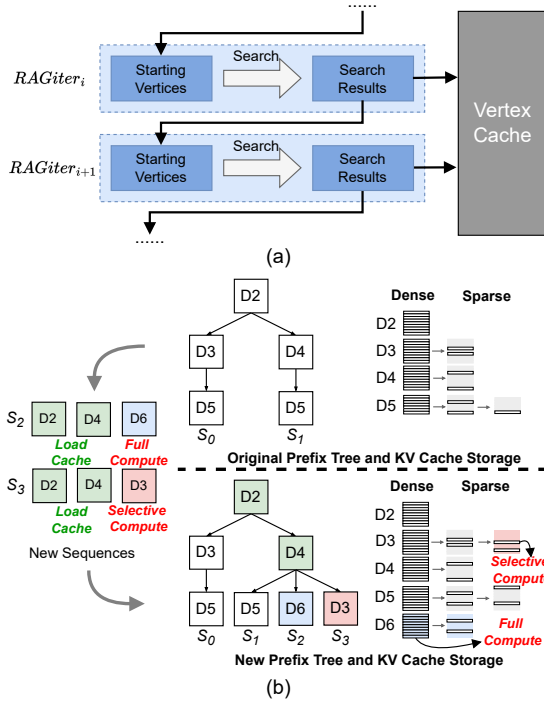


Figure 8: (a) The design of locality-aware retrieval optimization and (b) The processing of new document sequences using locality-aware generation

To address these challenges, we propose three software-hardware co-optimizations: locality-aware retrieval, locality-aware generation, and fine-grained parallel pipeline, which respectively improve the efficiency of the retrieval stage, the generation stage, and the overall system.

Locality-aware Retrieval: Figure 8(a) illustrates the design of locality-aware retrieval. This optimization involves two key approaches: first, since certain vertex vectors are frequently accessed during retrieval, we collect the results of each retrieval and cache frequently used vertex vectors to reduce DRAM accesses. Second, in iterative RAG scenarios, different iterations of the same user query often retrieve the same documents. To accelerate the search process, we reuse search results from the previous iteration as starting vertices for subsequent iterations.

To support locality-aware retrieval, we include a vertex vector cache in each RPM, as shown in Figure 6(e). To simplify the hardware design, all vertex vectors read from DRAM are directly used to update the cache, managed with the *Least Recently Used* (LRU) replacement policy. In iterative RAG scenarios, after retrieval, the vertex vectors in the result buffer are moved to the initial buffer to serve as starting vertices for the next iteration. For new user queries, the host resets the initial buffer. To optimize cache usage, the cache is divided into two parts: local and global. During query processing, updates are made only to the local cache. After the query is completed, the local cache is merged into the global cache.

Locality-aware Generation: We focus on the *prefilling* phase in the generation stage, where KV tensors corresponding to frequently retrieved documents are often recomputed multiple times.

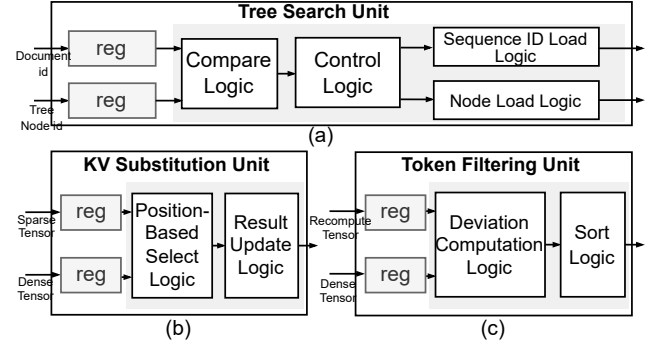


Figure 9: Overview of hardware units for locality-aware generation: (a) tree search unit, (b) KV substitution unit, and (c) token filtering unit

Caching KV tensors for each document consumes significant memory, as multiple copies may be needed for the same document. For instance, sequences like {D1, D2, D3} and {D1, D4, D3} require storing D3's KV twice. While prefix trees [35, 88] have been employed to share KV caches among sequences with common prefixes, this approach becomes less effective for long sequences with limited prefix overlap. Furthermore, KV is often recomputed when a document appears in a new sequence, even if its entries have already been cached.

Inspired by a recent study [87], we propose a method combining prefix trees with selective computation to improve caching and efficiency while maintaining generation quality. Documents are organized into a prefix tree, with nodes storing their tokens and KV addresses. KV is divided into two parts: dense and sparse. The dense part is computed based on the document itself, independent of any sequence. The sparse part keeps only the KV related to certain important tokens when the document appears in a sequence. Due to attention sparsity [9, 57], keeping only these important tokens does not reduce generation quality and greatly reduces storage for documents appearing multiple times in the tree. Selecting 10–20% of tokens typically causes just a 0.2% attention deviation, which has little effect on generation quality [87].

Figure 8(b) illustrates an example of processing new document sequences. After processing the sequences {D2, D3, D5} and {D2, D4, D5}, the system has built a prefix tree. For a new sequence {D2, D4, D6}, the system searches the tree and finds a match for {D2, D4}, but not for D6. For the matched documents D2 and D4, the system loads their dense and sparse KV tensors and constructs a new KV by selectively replacing parts of the dense KV with the sparse KV. Since document D6 does not match any existing path, its KV is not cached and will be computed during the next *prefilling* phase. When another sequence {D2, D4, D3} arrives, the same procedure is repeated. Although D3 does not match any existing path in the prefix tree, its dense KV exists in the cache, enabling selective recomputation of important tokens. The result is then cached as D3's new sparse KV. When the cache becomes full, the system evicts sparse KV entries based on the LRU replacement policy. When a document's sparse KV entries are all evicted, the system reclaims its cache space entirely.

To support locality-aware generation optimization, we add three customized hardware units to the TPM of the AccelHBM device: the

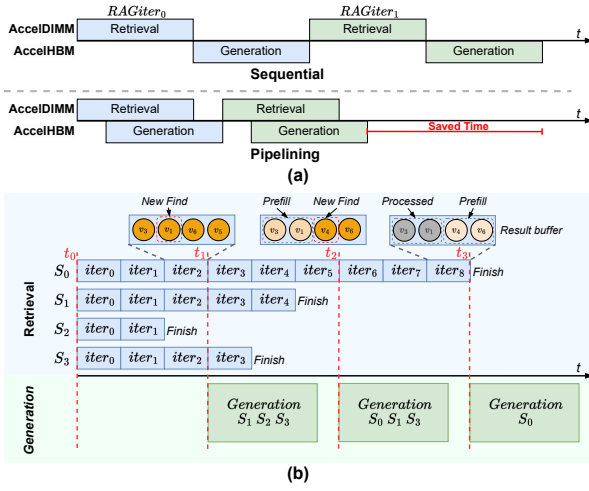


Figure 10: (a) Performance improvement opportunity through retrieval-generation overlap and (b) Fine-grained parallel pipeline optimization example

tree search unit, the KV substitution unit, and the token filtering unit. The design of the three hardware units is illustrated in Figure 9. The tree search unit matches new sequences in the prefix tree. The KV substitution unit combines dense and sparse KV of matched documents. For unmatched documents with cached dense KV, the token filtering unit selects important tokens.

Fine-grained Parallel Pipeline: Within a batch, retrieval tasks vary in difficulty, causing uneven completion times. Most tasks find neighbors quickly, while some take longer. In general, closer neighbors within a task are easier to find, while distant ones take more time [49, 91]. We can start generation once easier tasks are done or difficult tasks are partially completed, instead of waiting for all retrieval to finish. This allows overlap between retrieval and generation to speed up the process, as shown in Figure 10(a).

Specifically, the host aggregates retrieval results at fixed intervals. Completed searches send their top- k documents to AccelHBM, while incomplete ones send the most promising results. With locality-aware generation, AccelHBM can begin *prefilling* immediately upon receiving partial results. Additionally, for requests where retrieval is fully completed, AccelHBM smoothly moves to the *decoding* phase.

Figure 10(b) illustrates a simple example. The system processes four search tasks concurrently. At t'_1 , the host aggregates the intermediate results: task S2 has completed, and all of its results are sent to AccelHBM. Meanwhile, S0, S1, and S3 are still in progress. Since S1 and S3 are close to completion, the host identifies high-confidence results that are unlikely to change and sends them ahead. In contrast, for S0, the newly discovered vertex $v1$ appears near the head of the result buffer, indicating the search is still in its early stage, and the host therefore skips its results. At t'_2 , S1 and S3 are complete, and their remaining results are sent to AccelHBM. For S0, vertex $v4$ is newly found and appears near the tail of the result buffer; therefore, the host decides to send $v3$ and $v1$, as they are closer to the head of the result buffer. Finally, at t'_3 , S0 finishes its search, and the host transmits the remaining $v4$ and $v6$.

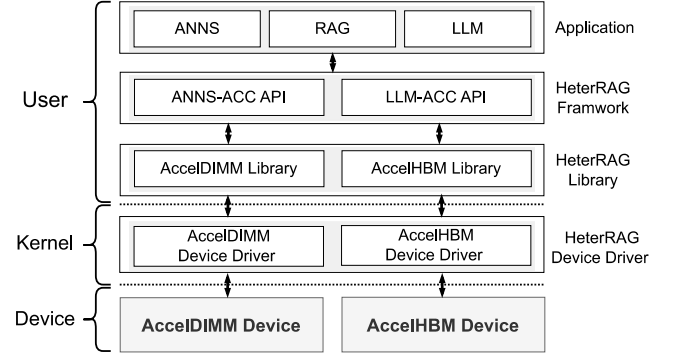


Figure 11: HeterRAG software stack

4.5 HeterRAG Software Stack

To simplify the use of HeterRAG, we develop a software stack based on the existing *Machine Learning* (ML) software stack. As shown in Figure 11, this stack extends the existing ML framework [1] with two primary *Application Programming Interfaces* (APIs): ANNS-ACC for ANNS and LLM-ACC for LLM inference. Together, these APIs enable the seamless execution of RAG workflows by leveraging the HeterRAG library to efficiently manage and coordinate devices like AccelDIMM and AccelHBM.

Programs written with the HeterRAG APIs are parsed by the HeterRAG compiler into intermediate representations, which are then compiled into host executables and device-specific binaries. These binaries are dispatched to devices via drivers. Upon receipt, the devices decode the instructions, execute the assigned tasks, and notify the host or return results directly.

4.6 Discussion

Generalization: First, HeterRAG's heterogeneous PIM architecture is well-suited for applications that demand both high memory bandwidth and large capacity, such as graph processing [11] and recommendation systems [39]. In these scenarios, data is typically divided into two categories: hub data, which is computationally intensive, and regular data, which is more abundant but less demanding [56, 85]. Based on this distinction, hub data is processed on HBM-based PIM, while regular data is handled by DIMM-based PIM, achieving an optimal balance between performance and cost. Second, HeterRAG supports multiple operations such as graph traversal, GEMM, and GEMV, which are fundamental to many applications, including graph neural networks [36].

HBM Memory Capacity: If the system needs to handle very large models, the HBM capacity may become insufficient. HeterRAG offers three solutions to address this issue, and users can choose any of them based on their specific needs: (i) leverage the scalability of HeterRAG by directly adding additional AccelHBM devices. Since generation tasks in practical RAG applications typically require much less memory than retrieval tasks, the cost of this approach is generally acceptable. Moreover, parallel inference strategies allow HeterRAG to deliver optimal performance; (ii) use standard DIMMs as extra memory, which is a lower-cost option but introduces data transfer bottlenecks between DIMMs and HBM-based PIM, potentially degrading performance; (iii) combine both approaches to achieve a balance between performance and cost.

Table 1: System configurations

Configuration of AccelDIMM	
DDR Specification	DDR4-3200 MT/s, 16Gb × 8, 2 Ranks
DDR Timing Parameters	tRC=72, tRCD=22, tCL=22, tRP=22, tBL=4, tRRD_S=4, tRRD_L=6, tFAW=30, tCCD_S=4, tCCD_L=6
Top-level Processing Module	128-entry Inst Queue, 1 Inst Decoder, 1KB Result Buffer, 16MB Visited List Buffer, 1KB Initial Buffer, 2KB Neighbor Buffer, and 128 FP32 Comparator × 8 for Top-k Priority Queue
In-memory Processing Module	256/128-entry PIM-Inst Queue for DPM/RPM, 1 PIM-Inst Decoder, 32KB Distance buffer for each DPM, 128KB Vertex Cache, 32 FP32 Mult, and 32 FP32 Adder for each RPM
Configuration of AccelHBM	
HBM Specification	8-Hi HBM2 stack, 8 Channels
HBM Timing Parameter	tRC=45, tRCD=16, tCL=16, tWR=16, tRAS=29, tRRD=2, tCCD_S=2, tCCD_L=4
Top-level Processing Module	128-entry Inst Queue, 1 Inst Decoder, 128×128 Systolic array × 8, 16 4-wide VLIW processors × 8, 1 TSU × 8, 1 KVSU × 8, 1 TFU × 8, and 24MB Scratchpad memory
In-memory Processing Module	32 FP16 Mult, 32 FP16 Adder, and 8Kb Buffer for each BPM

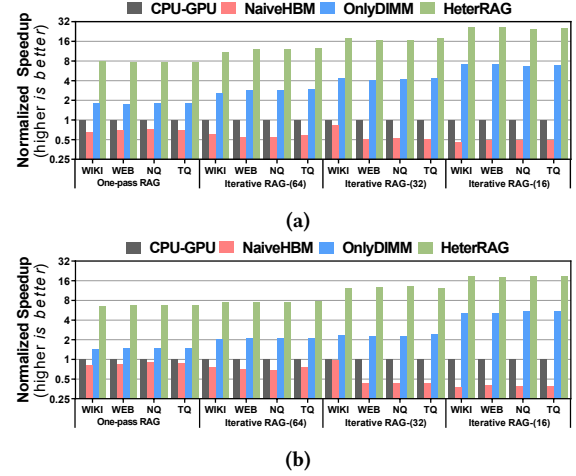
Table 2: The evaluated LLM configurations

Model	#Layers	#Head	d_{model}
GPT-2 XL-1.5B	48	25	1600
LLaMA2-7B	32	32	4096
LLaMA2-70B	80	64	8192

5 Evaluation

5.1 Experimental Setup

HeterRAG Settings: We develop a cycle-accurate simulation framework to comprehensively evaluate the performance of the HeterRAG system. Specifically, we extend Ramulator [41] to support top-level and in-memory processing modules, and modify ZSim [69] to offload the retrieval and generation of RAG to the memory side. For different DRAM types, we design two memory-side configurations: one based on DDR4 and the other on HBM2. The detailed system configurations are shown in Table 1. We implement the arithmetic units using Verilog HDL and synthesize them using Synopsys Design Compiler [73] with 65nm technology at 500MHz clock frequency. The power and area data of the in-memory arithmetic units are scaled to 22nm to match the memory technology, adding about 50% more area to account for differences between logic and DRAM processes. DDR4 power consumption is calculated using the MICRON DDR4 power calculator [60], while HBM2 power consumption is based on TPU-v4i [38]. Additionally,

**Figure 12: Normalized throughput of CPU-GPU, NaiveHBM, OnlyDIMM, and HeterRAG. We use the two models (a) GPT-2 XL and (b) LLaMA2-7B**

we use CACTI [5] to model the latency, energy consumption, and area of all buffers/caches in the system.

LLM Models: We select two commonly used open source language models for evaluation in RAG systems: GPT-2 [67] and LLaMA2 [75], with parameter sizes ranging from 1.5B to 70B. Specifically, LLaMA2-70B is used to evaluate scalability. Table 2 provides detailed model configurations. It is worth noting that HeterRAG is highly compatible and supports any language model based on the transformer decoder architecture.

Datasets and Knowledge Base: We choose four question answering datasets commonly used in knowledge-intensive open domains: *Wiki-QA* (WIKI) [86], *Web Questions* (WEB) [6], *Natural Questions* (NQ) [42], and *Trivia-QA* (TQ) [37]. To better assess HeterRAG’s generation capability, we manually extended the output length during inference. The knowledge base is constructed based on the Wikipedia corpus [10] and uses the graph-based ANNS algorithm HNSW [58] for retrieval.

Baselines: We compare HeterRAG with three baseline systems: (i) **CPU-GPU** is a real system equipped with an Intel Xeon Gold 5117 CPU (2.00GHz, 256GB DDR4) and an NVIDIA Tesla V100 GPU (32GB HBM2). Energy consumption is measured using Intel RAPL and nvprof; (ii) To validate the architecture’s effectiveness, we build the **NaiveHBM** baseline, where both retrieval and generation run on HBM-based PIM. NaiveHBM consists of two devices: AccelHBM-g and AccelHBM-r. AccelHBM-g is the same as in HeterRAG, while AccelHBM-r uses standard DIMMs and PIM-enabled HBM. For retrieval, the DPM is placed in the HBM buffer die, with one RPM assigned to each channel; (iii) **OnlyDIMM** runs both retrieval and generation on DIMM-based PIM, with each DIMM bank equipped with a BPM for generation tasks. All baselines use the same size for each type of DRAM. For instance, the HBM size in AccelHBM is identical to the V100 GPU.

5.2 Overall Performance

Throughput: Figure 12 compares the throughput of HeterRAG with CPU-GPU, NaiveHBM, and OnlyDIMM under different configurations. The results show that the NaiveHBM baseline performs

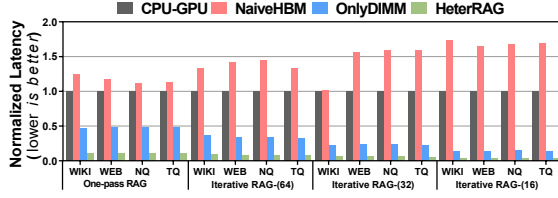


Figure 13: Normalized latency of CPU-GPU, NaiveHBM, OnlyDIMM, and HeterRAG on LLaMA2-7B model

Table 3: Average resource utilization of AccelDIMM and AccelHBM across all evaluation configurations

	HeterRAG-base	HeterRAG-p
AccelDIMM	37.8%	44.5%
AccelHBM	31.0%	38.6%

worse than the CPU-GPU baseline due to the overhead of data transfers from DIMMs to HBM, which offsets the performance gains from the PIM design. In contrast, the OnlyDIMM baseline performs better as it relies solely on DIMM-based PIM, achieving high bandwidth for both retrieval and generation tasks. As a result, OnlyDIMM improves throughput by $1.41\times$ to $7.21\times$ ($3.38\times$ on average) compared to the CPU-GPU baseline. HeterRAG further improves throughput by $6.62\times$ to $26.53\times$ ($13.52\times$ on average) over the CPU-GPU baseline.

Latency: Figure 13 compares the end-to-end latency of a single RAG request for HeterRAG and various baselines. The results show that HeterRAG processes each RAG request with exceptionally low latency. On average, HeterRAG reduces response latency by $13.72\times$, $19.47\times$, and $4.07\times$ compared to CPU-GPU, NaiveHBM, and OnlyDIMM, respectively.

Energy Efficiency: HeterRAG achieves higher energy efficiency than the CPU-GPU, NaiveHBM, and OnlyDIMM baselines, as shown in Figure 14(a). Compared to the CPU-GPU baseline, HeterRAG reduces energy consumption by 35.50% to 63.76% (49.8% on average), owing to its PIM design, which substantially reduces the energy required for memory access. Compared to the NaiveHBM baseline, HeterRAG reduces energy consumption by an average of 17.0%. Against the OnlyDIMM baseline, HeterRAG lowers energy consumption by an average of 72.25%, taking advantage of HBM’s low energy per bit transfer. Figure 14(b) presents the energy breakdown across different baselines, showing that HeterRAG reduces energy consumption in retrieval, generation, and data transfer due to its efficient architecture and software-hardware optimizations.

5.3 Implication of Optimization

For HeterRAG, we evaluate multiple variants to analyze the impact of the three proposed optimizations. Specifically, HeterRAG-base has no optimizations, HeterRAG-r applies locality-aware optimization for the retrieval stage, HeterRAG-g applies locality-aware optimization for the generation stage, HeterRAG-p introduces fine-grained parallel pipeline optimization, and HeterRAG-rgp combines all optimizations. Figure 15 compares the throughput performance of the five different HeterRAG variants.

Locality-aware retrieval, locality-aware generation, and fine-grained parallel pipeline optimization improve throughput by an average of $1.25\times$, $1.22\times$, and $1.20\times$, respectively. Locality-aware

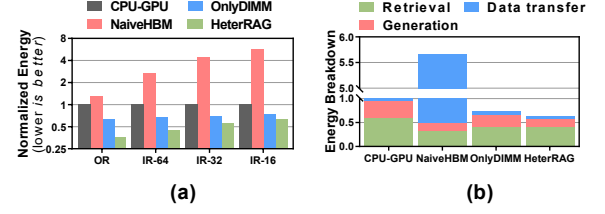


Figure 14: (a) Normalized energy and (b) Energy breakdown per request for CPU-GPU, NaiveHBM, OnlyDIMM, and HeterRAG (LLaMA2-7B, results averaged across four datasets)

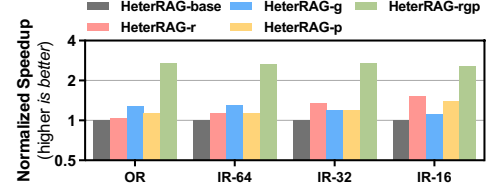


Figure 15: Normalized throughput of HeterRAG variants (LLaMA2-7B, results averaged across four datasets)

generation shows the most improvement in configurations where the generation stage dominates (e.g., one-pass RAG), while locality-aware retrieval is more effective when the retrieval stage dominates (e.g., iterative RAG with a retrieval interval of 16). Fine-grained parallel pipeline optimization provides consistent benefits across all configurations. As shown in Table 3, it enhances resource utilization for both device types to 44.5% and 38.6%, demonstrating its effectiveness. These results demonstrate the adaptability of the three optimization techniques, allowing users to choose and combine them based on specific needs.

5.4 Scalability

With the growing scale of data, it is important to evaluate the scalability of HeterRAG when adding heterogeneous PIM devices. We analyze this from two aspects: retrieval and generation.

Retrieval: We synthesize a larger Wikipedia knowledge base with approximately 1 billion entries, exceeding 4TB in size, and increase the number of AccelDIMM devices from 4 to 32 to measure retrieval throughput. As shown in Figure 16(a), HeterRAG achieves near-superlinear throughput improvement. This is due to data parallelism, where adding more AccelDIMMs reduces the workload per device. With no inter-device communication, only a small amount of results is sent to the host for aggregation.

Generation: We use the LLaMA2-70B model with each AccelHBM configured to 80GB HBM capacity. We increase the number of AccelHBMs from 4 to 10 to measure inference throughput. As shown in Figure 16(b), a $2.5\times$ increase in AccelHBMs leads to a $1.78\times$ improvement in throughput, showing decent scalability. However, linear speedup is limited because the generation phase involves unavoidable communication between devices.

5.5 Comparison with Prior Work

Retrieval: We compare HeterRAG with MemANNS [15] and DRIM-ANN [14], both using the UPMEM module, and evaluate them on the SIFT dataset. As shown in Figure 17(a), HeterRAG achieves an average $25.15\times$ and $28.42\times$ higher *Queries Per Second* (QPS), respectively. This performance gain comes from two factors: HeterRAG’s

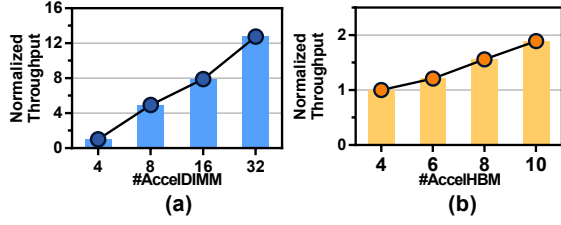


Figure 16: Scalability of (a) AccelDIMM and (b) AccelHBM

graph-based ANNS is more efficient than the PQ-based ANNS used by MemANNS and DRIM-ANN, and its specialized hardware outperforms the general-purpose RISC cores of UPMEM.

Generation: We compare HeterRAG with two state-of-the-art HBM-based PIM acceleration approaches on the LLaMA2-70B model. As shown in Figure 17(b), HeterRAG exhibits performance comparable to NeuPIMs [26] and AttAcc [64].

End-to-end RAG: We compare HeterRAG with two existing heterogeneous RAG acceleration solutions, Chameleon [33] and PipeRAG [34], using the same benchmark as reported in the respective papers. As shown in Figure 17(c) and (d), with architectural and algorithmic innovations, HeterRAG achieves an average $6.17\times$ higher throughput than FPGA-GPU-based Chameleon and $5.48\times$ lower latency than CPU-GPU-based PipeRAG.

5.6 Area Overhead

We estimate the area overhead of the two PIM designs in HeterRAG. For HBM, one BPM is assigned to every two banks, adding a total area overhead of $6.016mm^2$ on the DRAM die, which is about 11.31% of the $53.15mm^2$ HBM2 [93]. Most of this comes from the arithmetic units. For DIMM, each RPM requires about $3.368mm^2$, and each DPM adds $0.842mm^2$, totaling $7.817mm^2$ per DIMM. Since a typical buffer chip area is over $100mm^2$, this overhead is acceptable.

6 Related Work

Approximate Nearest Neighbor Search Acceleration: A lot of research has focused on optimizing ANNS, including hash-based [71], tree-based [18, 22], and product quantization-based [4] methods. Recently, graph-based [19, 58, 82] ANNS methods have become the mainstream in academia and industry due to their strong performance in search tasks. On the hardware side, many studies have explored using GPUs [63], FPGAs [89], and ASIC-based [47] hardware to accelerate ANNS. Unlike these studies, our work focuses on in-memory acceleration for ANNS in RAG scenarios and investigates how to leverage locality in RAG to improve performance.

Large Language Models Acceleration : Many studies accelerate model inference through software-hardware co-design. For example, A3 [24] proposed an approximate attention mechanism with dedicated hardware. Although these methods improve performance, they do not address the memory bottleneck. Recently, some studies have explored PIM techniques to speed up transformer-based LLMs, including TransPIM [93], AttAcc [64], NeuPIMs [26], and IANUS [70]. Others combine algorithm optimization with PIM, such as PIM-DL [50], which replaces GEMM computations with lookup table operations for better in-memory performance, and SpecPIM [51], which uses PIM to accelerate speculative inference. Unlike these studies, our work focuses on identifying the best PIM

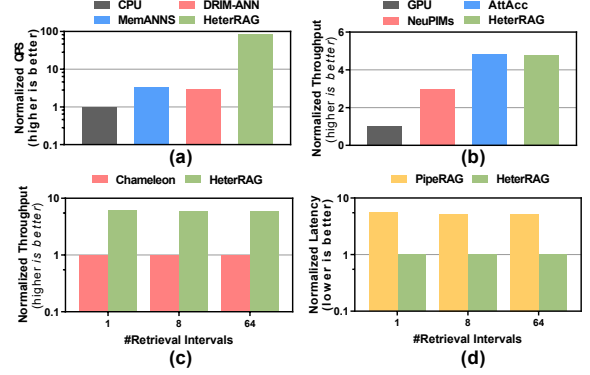


Figure 17: Performance comparison of HeterRAG with (a) MemANNS and DRIM-ANN, (b) NeuPIMs and AttAcc, (c) Chameleon, and (d) PipeRAG

design for model inference in RAG systems and optimizing for RAG-specific features.

Retrieval-augmented Generation Acceleration: Some recent studies focus on optimizing RAG systems. Some focus on software optimizations, such as RaLMSpec [92], which improves RAG's retrieval stage using speculative retrieval mechanisms, and RAG-cache [35], which primarily optimizes the generation stage with the knowledge tree structure. PipeRAG [34] aggressively reuses retrieval results from previous iterations for the current generation stage, improving overall system performance. Unlike these methods, our optimizations consider retrieval, generation, and pipeline comprehensively, with a design distinct from existing works. Combining ideas from these studies could further improve the performance of HeterRAG. On the hardware side, Chameleon [33] uses a heterogeneous FPGA and GPU design to accelerate RAG. However, it relies on PQ-based ANNS for retrieval and does not address the memory bottleneck in RAG systems.

7 Conclusion

Retrieval-augmented Generation (RAG) is widely used and involves two stages: retrieval and generation. However, it faces challenges due to high memory capacity and bandwidth demands. Existing HBM-based PIM solutions struggle to meet both requirements simultaneously. To address this, we present HeterRAG, a heterogeneous PIM system combining HBM-based PIM and DIMM-based PIM. DIMM-based PIM provides larger memory capacity, complementing HBM-based PIM while ensuring high performance, energy efficiency, and lower hardware costs. We also propose three optimizations to improve retrieval and generation efficiency and enhance pipeline execution. Evaluation results show that HeterRAG significantly outperforms the CPU-GPU baseline in throughput, latency, and energy efficiency.

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